The Hidden Danger: Insider Threats in Industrial Control Systems in 2025

By **Matthew Schaefer** (https://westoahu.hawaii.edu/cyber/author/mschaefe/) on February 28, 2025

Executive Summary

Insider threats in Industrial Control Systems (ICS) are not just a possibility, instead they're a growing problem. Employees, contractors, and third-party vendors with authorized access can, whether intentionally or not, create major security risks. The given risks can go beyond just data breaches, insider incidents can disrupt operations, cost millions, and even put public safety in jeopardy. A study from DTEX Systems citing Ponemon's Cost of Insider Risks Report found that 83% of organizations faced at least one insider security breach last year. These aren't just minor disruptions. The financial consequences alone can be staggering, with companies losing on average $15.4 million per incident. As these threats become more sophisticated in 2025, organizations can't afford to be reactive [1].

Background

Unlike external cyberattacks that require breaching perimeter defenses, insider threats originate from trusted individuals within an organization. Malicious insiders intentionally sabotage systems, leak sensitive information, or manipulate control settings for financial gain or personal grievances. In contrast, negligent insiders unknowingly expose vulnerabilities by falling victim to phishing attacks or failing to follow security protocols. The convergence of Information Technology (IT) and Operational Technology (OT) in Industrial Control Systems (ICS) environments has further increased exposure, making insider threat detection more complex [2]. Not all insider threats look the same. Some are deliberate acts of sabotage or data theft, while most are often not but rather unintentional mistakes. According to DTEX Systems, 55% of insider related incidents in ICS environments are caused by negligence. These are simple missteps like clicking a phishing link or misconfigured security settings. Meanwhile 26% are the result of malicious insiders who abuse their access for financial gain or personal revenge. The shift to remote work and cloud-based industrial operations have expanded access points, increasing the risk of unauthorized system modifications [3]. Recent reports highlight that over 30% of ICS-related cybersecurity incidents in the past year involved insider actions, emphasizing the urgency of proactive threat mitigation [4].

Impact

The damage insider threats can cause are massive. If you imagine a single compromised account leading to shutdowns in energy plants or water treatment facilities. Then it goes beyond being just an inconvenience. It can create real public safety issues. Malicious insiders can exploit their access to disable security systems, introduce malware, or even disrupt critical operations. In many cases, these incidents could have been prevented with stronger monitoring and controls. For instance, a disgruntled former employee at a power plant recently attempted to alter load distribution settings, risking widespread blackouts. Another case involved a contractor unintentionally introducing ransomware into an ICS network, halting production for weeks. The numbers do paint a troubling picture. Insider caused security failures in ICS environments have jumped by 35%, according to DTEX Systems. This spike wouldn't be random, it;s a result of insufficient monitoring, weak access controls, and a lack of behavioral analytics. Without better detection mechanisms, organizations will continue to suffer costly breaches that could all be prevented. Financial repercussions from such incidents include forensic investigations, recovery costs, regulatory penalties, and reputational damage. Beyond monetary loss, public trust in critical infrastructure operations diminishes when insider threats compromise service reliability.

Mitigation

Organizations need to step up their game within layered security defenses. This means not just stronger passwords and firewalls, but introducing your system to smarter monitoring tools. Such examples can be behavioral analytics and AI powered sensors to spot the unusual activity, like unauthorized data transfers or odd login times. Organizations that fail to invest in these solutions are already falling behind. A 35% rise in insider caused failures tracked by DTEX Systems, proves that traditional security approaches aren't enough. Without the behavioral analytics and continuous monitoring, insider threats will continue to slip through the cracks. Enforcing zero-trust architecture ensures that all users and devices must continuously verify their identity, preventing unauthorized lateral movement within networks. Mandatory employee training

programs should educate personnel on insider threat indicators, phishing scams, and best security practices. Additionally, organizations must limit third-party access and require security assessments for vendors handling ICS components. By implementing continuous monitoring, strict privilege controls, and real-time alerts, ICS operators can reduce insider threat risks and enhance overall security resilience.

Relevance

As ICS security threats evolve, insider threats demand immediate attention. Most cybersecurity defenses are built to stop outside attacks. When these threats come from within, they bypass traditional security measures, making detection far more difficult. That's what makes insider attacks so dangerous as they can often slip past the traditional security measures that may go unnoticed. Companies that don't take insider threats seriously aren't just risking financial losses, they're setting themselves up for disruptions that could grind operations to a halt or an end. By prioritizing a mix of zero-trust security, behavioral monitoring, access restrictions, and advanced threat detection. These ICS operators can ensure long-term security and reliability in an increasingly complex threat landscape.

References

[1] Roessler, K. (2025, Feb 26). *2025 Ponemon Cost of Insider Risks Global Report.* DTEX Systems. https://www.dtexsystems.com/blog/2025-cost-insider-risks-takeaways/ (https://www.dtexsystems.com/blog/2025-cost-insider-risks-takeaways/)

[2] Ribeiro, A. (2023, April 30). *Evaluating ICS Cyber Threat Landscape Focusing on Insider Threat in OT Environments.* Industrial Cyber. https://industrialcyber.co/features/evaluating-ics-cyber-threat-landscape-focusing-on-insider-threats-in-ot-environments (https://industrialcyber.co/features/evaluating-ics-cyber-threat-landscape-focusing-on-insider-threats-in-ot-environments)

[3] NanoLock Security. (2024, September 24). *Protecting Industrial Devices: The Rising Threat of Insider Attacks.* Nanolock Security. https://nanolocksecurity.com/protecting-industrial-devices-the-rising-threat-of-insider-attacks (https://nanolocksecurity.com/protecting-industrial-devices-the-rising-threat-of-insider-attacks)

[4] Lota, S. (2024, November 21). *Combating Insider Threats in OT & ICS Environments.* Nozomi Networks. https://www.nozominetworks.com/blog/insider-threats-in-ot-ics-call-for-endpoint-sensors-and-behavioral-analytics (https://www.nozominetworks.com/blog/insider-threats-in-ot-ics-call-for-endpoint-sensors-and-behavioral-analytics)

RELATED POSTS

Malicious Packages Targeting ICS Software (https://westoahu.hawaii.edu/cyber/ics-cybersecurity/ics-weekly-summaries/malicious-packages-targeting-ics-software/)
11/18/2025

British Water Facilities Breached Through ICS Exposure's (https://westoahu.hawaii.edu/cyber/ics-cybersecurity/ics-weekly-summaries/british-water-facilities-breached-through-ics-exposures/)
11/12/2025

Several Canadian ICS Systems Breached (https://westoahu.hawaii.edu/cyber/ics-cybersecurity/ics-weekly-summaries/several-canadian-ics-systems-breached/)
11/4/2025